

• BITTENSOR SUBNET · COMPUTER-USE GROUNDING Bittensor子网 · 计算机使用接地



A subnet that makes computer-use agents accurate enough to ship.

让计算机使用代理足够精准、可以投产的子网。

We pay open-weight models for one thing only — turning a natural-language instruction and a screen into the **right pixel**. Validators score it. Emissions follow accuracy.

我们只为开源权重模型做一件事付费——把自然语言指令和屏幕变成**正确的像素**。验证器打分，奖励跟随准确率。

01 / 11 · COVER

SUBNET

DDX · sn candidate

MECHANISM

Yuma **trimmed-mean**

stake-weighted consensus

BENCHMARKS

UI-Vision · OSWorld-G · SS-Pro

TIME TO LIVE DEMO

48 hours

TEAM

Drew Mimi

Keith Guo

GROUND TRUTH, ON-CHAIN.

- THESIS 论点

Computer-use agents don't fail at reasoning. They fail at finding the pixel.

计算机使用代理不是输在推理上，而是输在找像素上。

66%

Agent failures traced to grounding errors.

代理失败可追溯到接地错误。

Reasoning is solved-enough. The model picks the wrong coordinate, the

\$139-199B

Agentic AI infrastructure by 2034 (34%+ CAGR).
到 2034 年的代理型 AI 基础设施。

Anthropic, OpenAI, Google all train grounding directly. Frontier hits 86% on

ScreenSpot-Pro — saturating. **UI-Vision** and fine-grained **OSWorld-G**

remain **below 40%** end-to-end. Specialist open models (UI-TARS, OS-

0

Open networks paying miners to win GUI grounding.

为矿工赢得 GUI 接地付费的开放网络。 THE BOTTLENECK MOVED.

Open-source teams ship in isolation. No coordinated incentive. DDX is the

A defensible scoring mechanism.

经得起推敲的评分机制。

PER-PREDICTION SCORE
单次预测得分

```
// for miner m, prompt p:
score(m, p)
= IoU(predm,p, gtp)
× 1[IoU > τ]
× exp(-λ · latency)
× robustness(m, p)
```

PER-TEMPO EMISSION WEIGHT
单节奏奖励权重

$$W_m = \mu(\text{score}_m, b)^k / \sum_j \mu(\text{score}_j, b)^k$$

// k = 3 · power-law concentration on frontier

IoU

Intersection-over-union of predicted & ground-truth bbox. **IoU > 0.5 is the stricter variant of click-within-bbox we chose deliberately.**
预测与真实边框的交集面积比。严于点击命中中的变体。

t

Hard gate at **0.5**. No partial credit for “close”.
硬门限 0.5, “接近” 不得分。

λ

Latency soft target **300ms**; decays exponentially, hard ceiling at **1500ms**.
300ms 内满分; 1500ms 处为零。

robustness

Worst-case score across resolution / theme / font-scaling variants of the same prompt.
同一提示词在多种屏幕变体下的最坏情况得分。

k = 3

Power-law exponent. Concentrates emissions on the frontier while preserving 2nd / 3rd place economics.
幂律指数, 集中奖励前沿。

Four components — IoU, a hard gate, a latency penalty, robustness across screen variants. Nothing learned, nothing tunable per-miner. 200 lines of validator code; weeks of adversarial review.

四个组成部分——IoU、硬门限、延迟惩罚、跨屏幕变体的健壮性。无可学习、无可调参。200 行验证器代码；数周对抗审查。

Three benchmarks. One weighted score.

三个基准。一个加权得分。

Public, instrumented, peer-reviewed. We blend them so no single benchmark dominates — and so miners that overfit to one lose ground on the composite.

UI-V

weight0.40

UI-Vision

桌面端 · 细粒度接地
Nayak et al. 2025 · 83 desktop applications

Desktop-centric, license-permissive, fine-grained. Element grounding, layout grounding, action prediction. Best end-to-end published score: **36.2%** (Phi-Ground). Where real desktop agents still fail.

APPS	TASKS
83	7,500+
SCORE TYPE	CEILING
click-in-bbox	<40% · headroom

OSW-G

weight0.30

OSWorld-G

NeurIPS 2025 Spotlight · 精确控件
Xie et al. 2025 · NeurIPS Spotlight

Fine-grained UI manipulation across 32 distinct element types. Tests precise widget interaction that other benchmarks miss. Open-source SOTA: **Qwen3-VL-235B at 68.3%**.

ELEMENT TYPES	TASKS
32	564
SCORE TYPE	CEILING
click-in-bbox	<70% · headroom

SS-PRO

weight0.30

ScreenSpot-Pro

专业软件 · 4K · 对比基线
Li et al. 2025 · 1,581 high-res instances

Professional software — Photoshop, AutoCAD, Blender, Premiere. 4K resolution, dense UI. Established leaderboard. Frontier: **GPT-5.2 86.3%**. Open-source: **UI-Venus-72B 61.9%**.

APPS	RESOLUTION
23	3840×2160
SCORE TYPE	OPEN SOTA
IoU	62% · less room

UPCOMING v1.1 adds **VisualWebArena fragments** (0.10) and **a private DDX-generated set** (0.10) once 50 validators are live. Weights are governed by stake vote.

VALIDATOR UI · LIVE · VALI-04.DDX

Anyone can audit the loop, live.

任何人都能实时审计这个循环。

Public read-only validator console. Real-time leaderboard, score history, anti-gaming flags, batch rotation status. Every value is verifiable against on-chain commits.

Mockup represents target operational state. Real numbers visible at vali-04.ddx.network once testnet is live.

VALIDATOR

●

vali-04.ddx

EPOCH

8,421

BATCH SIZE

256 prompts

ACTIVE MINERS

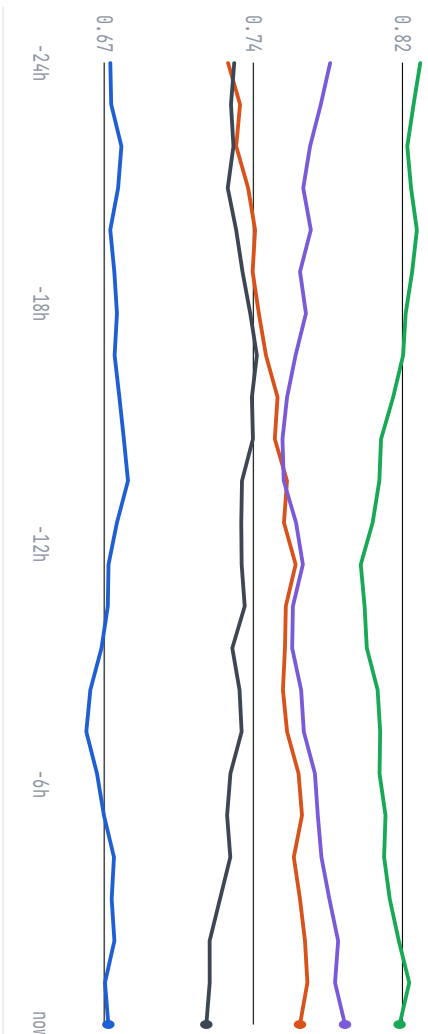
187 / 256

VALIDATOR LEADERBOARD · LIVE

updated 0s ago

#	UID · HOTKEY	MINER	UI-V	SS-PRO	OSW-G	WEIGHTED	A24H	24H TREND	EMISSION	
1	● 47	567c5...ef4d	shred-v3	0.848	0.621	0.583	0.702	+0.012		13.7%
2	● 12	57c52...f4da	groundwork	0.820	0.604	0.602	0.690	+0.004		12.4%
3	● 89	581ef...0967	tarsier-os	0.814	0.598	0.561	0.675	+0.018		11.6%
4	● 33	50967...81ef	pixelfoundry	0.795	0.572	0.547	0.655	-0.003		10.8%
5	● 64	5b096...381e	atlas-fine	0.790	0.589	0.498	0.647	+0.007		9.2%
6	● 21	54dab...c523	showui-lab	0.762	0.541	0.521	0.624	-0.005		8.4%
7	● 58	5dab0...5238	kicks	0.821	0.412	0.508	0.600	-0.041		6.1%
8	● 3	5ab09...2381	uitars-tiny	0.735	0.510	0.486	0.594	+0.002		6.4%
9	● 91	52381...ab09	click-and-see	0.702	0.488	0.451	0.564	-0.001		5.2%
10	● 17	50967...81ef	bbox-bandit	0.683	0.461	0.422	0.540	+0.005		4.4%

TOP 5 · WEIGHTED SCORE · LAST 24h



ANTI-GAMING SURFACE · LAST 6M

02:31:44

58

Held-out overlap detected on ScreenSpot-Pro subset

02:30:11

17

Submission latency > 40 from miner baseline

02:28:55

89

Validator quorum: 14 / 16 confirms

02:27:02

47

Reproducibility seed match: 100 / 100

02:25:39

33

Ensemble-via-UID flag cleared after review

NEXT SET ROTATION

73 blocks

TOTAL SUBMISSIONS

75,935

DEMO · GROUNDING IN ACTION

Prompt → bbox, scored live.

提示词 → 边界框，实时打分。

PROMPT

"Click the unread filter in Gmail's inbox toolbar"

Scene

Type

Source

gmail

natural-language → bbox

UI-Vision held-out

QUEUE · 1 / 4

01 Click the unread filter in Gmail's inbox toolbar

02 Find the Send button at the bottom of the compos...

03 Click the merge button on this pull request page

04 Open the Settings menu in this app's sidebar

SCREEN CAPTURE · 1920 × 1080

Gmail

Search mail

Compose

Inbox 2,341

Starred

Snoozed

Sent

Drafts 4

More

All

Unread

Starred

1-50 of 2,341

Stripe receipts

Caitlin via Linear

alex@anthropic.com

TAO holders newsletter

Subnet ops weekly

Drew Mirni

GitHub

Calendar

Notion

Discord

Your invoice is ready — please review

RE: Subnet emission curve discussion

RE: Subnet emission curve discussion

RE: Subnet emission curve discussion

RE: Subnet emission curve discussion

Your invoice is ready — please review

RE: Subnet emission curve discussion

10:00

11:07

12:14

13:21

14:28

15:35

16:42

17:49

18:56

19:03

FRAME · GMAIL

103, 243 → 206, 280

PREDICTION

IOU VS GROUND TRUTH

0.812

LATENCY

142 ms

ROBUSTNESS

0.859

GATE PASSED

true

CONFIDENCE

91.3%

tie-breaking only

shred-v3 / uid 47

subnet candidate · validator vali-04.ddx

batch hash · 0xa3f7,,91c2

Yuma trimmed mean across validators

// scoring

score = IoU(pred, gt)

× 1[IoU > 0.5]

× exp(-λ · latency)

× robustness(m, p)

We are not inventing the mechanism. We are porting it.

我们没有发明机制，我们在移植它。

DEPLOYED PRECEDENT · SCORE VISION (SN44)

Real subnet. Real ARR. Same submission-and-scoring pattern.

真实部署。真实营收。同样的提交与评分模式。

Score Vision is a deployed Bittensor vision subnet running a submission-and-scoring marketplace. Different domain (sports video analysis), same architectural pattern as DDX. Proof that decentralized vision marketplaces convert technical progress into enterprise revenue.

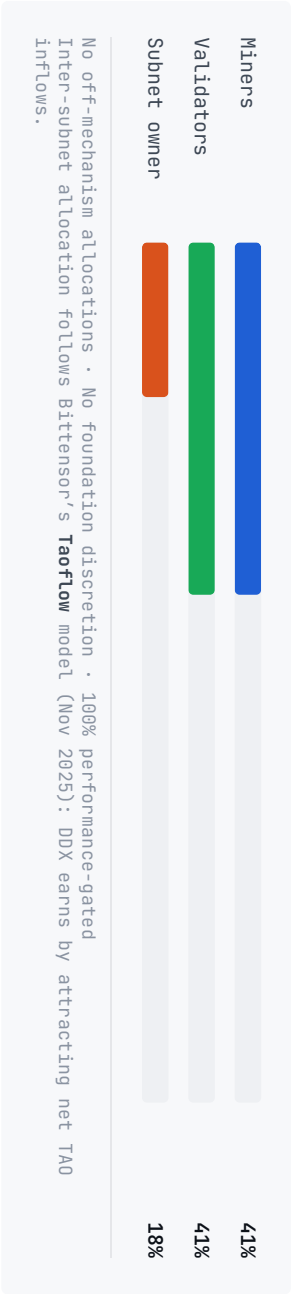
LIVE SINCE 2024	ENTERPRISE ARR ~\$3M
ENTERPRISE CLIENTS 7	AVG CONTRACT \$428K

EMISSIONS ALLOCATION · DDX

Where TAO goes.

TAO 奖励去哪里。

Standard 18 / 41 / 41 split between validators, miners, and subnet owner. Miner share routes 100% to held-out accuracy. No off-mechanism payouts, no “contribution” bonuses, no foundation discretion.



Standard Bittensor consensus: stake-weighted Yuma trimmed mean, power-law emissions, sealed held-out scoring. The commercial pattern is proven by **Score Vision (SN44)**, a deployed vision subnet with real enterprise ARR.

标准 Bittensor 共识：质量加权的 Yuma 修剪均值、幂律奖励、密封保留评分。商业模式由已部署的 Score Vision (SN44) 验证。

- 12-WEEK ROADMAP

Hackathon → testnet → mainnet → ship.

黑客松 → 测试网 → 主网 → 上线。

NOW WEEK 0 · HACKATHON May 23 – May 26	PHASE 1 TESTNET Week 1 – 4 Stand up testnet. 建立测试网。 - Validate scoring loop against UI-Vision, OSWorld-G held-out batches - Add robustness multiplier + distribution-shift batch to validator - Recruit 8 testnet validators (Score Vision ops, friends) - Onboard 25 testnet miners with starter weights - Adversarial review of anti-gaming surface	PHASE 2 MAINNET Week 5 – 8 Register sn-32. Open emissions. 注册 sn-32。开启奖励。 - Submit subnet registration. Accept the next available slot. - Migrate held-out pool from testnet - Stake-weighted Yuma trimmed-mean consensus live - Public leaderboard launch	PHASE 3 ECOSYSTEM Week 9 – 12 Ship the API. 发布 API。 - HTTP grounding API gated to top-5 miners - First 3 paying users (agent frameworks) - Adopt Score Vision (SN44) integration playbook for enterprise sales - v1.1 weight vote: VisualWebArena, DDX-private 50-validator decentralization milestone
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Submission ready.

提交就绪。

- Validator binary v0.3 with ScreenSpot scoring loop
- Live dashboard at vali-O4.ddx.network
- Baseline miner reproducing UI-TARS-7B
- This deck + 4-min stage demo

- TEAM & ASK

Two operators. One mechanism. Shipping in 48 hours.

两个执行者。一套机制。48 小时内上线。

MECHANISM • VALIDATOR

机制 • 验证器

Drew Mimi

Owens the validator binary, the scoring function, and the anti-gaming surface. Writing every line of consensus-critical code with adversarial review built in.

MODELS • MINER STACK

模型 • 矿工栈

Keith Guo

Owens the baseline miner, the training data pipeline, and the open-weights release cadence. Shipping a competitive grounding miner benchmarked against **UI-TARS-72B** and **UI-Venus-72B** on day one.

Demo on stage • validator console
drew@ddx.network

Demo on stage • grounding model
keith@ddx.network

ASK • WHAT WE NEED FROM THIS HACKATHON

A subnet slot, validators, and miners.

一个子网名额、验证器和矿工。

- Bittensor Foundation — subnet registration green-light (any open slot)
- Score Vision ops (or another deployed vision subnet operator) — adversarial review of scoring loop
- 4 launch validators with >500 TAO stake
- Intros to UI-TARS, OS-Atlas, UI-Venus teams as launch miners
- \$50k to fund 8 weeks of validator infra + adversarial audit